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PROFESSIONAL SUMMARY

M.S. candidate in Statistical Science at Indiana University Bloomington with expertise in Bayesian modeling, statistical learning, and scientific computing, targeting industry statistician roles. Develops and benchmarks scalable research software in R, C++, and Python, with hands-on experience accelerating computational pipelines and applying predictive modeling methods to real data.

EDUCATION

Indiana University, College of Arts and Sciences

Bloomington, IN

Master of Science in Statistical Science

Expected May 2027

Thesis: Time-Series Analysis Using Bayesian Additive Regression Trees (BART)

Graduate Coursework: Introduction to Statistical Computing; Applied Statistical Computing; Fundamentals of Statistics I–II; Applied Linear Models I–II

Current Coursework: Statistical Consulting; Research in Statistics; Introduction to Analysis I

University of Dhaka

Dhaka, Bangladesh

Bachelor of Science in Physics

August 2024

EXPERIENCE

Research Assistant, StatMIND Lab

2025 – Present

Indiana University Bloomington

Bloomington, IN

- Develop computational methods to improve the efficiency and scalability of Bayesian neuroimaging pipelines using R, and C++.
- Achieved a 4 to 5× speedup in Bayesian neuroimaging algorithms and benchmarked performance using simulated and real fMRI data.
- Build reproducible benchmarking workflows to compare runtime, scalability, and computational performance across implementations.

Teaching Assistant, Department of Statistics

2025 – Present

Indiana University Bloomington

Bloomington, IN

- Support undergraduate Statistics courses through classroom instruction, discussion sections, and individualized guidance.
- Evaluate approximately 200 assignments per grading cycle and provide detailed feedback on statistical reasoning, computation, and interpretation.
- Conduct weekly office hours and exam-review sessions to strengthen students' conceptual understanding and problem-solving skills.

Senior Mathematics Instructor

2016 – 2025

Udyash Academic & Admission Care

Dhaka, Bangladesh

- Taught and mentored more than 100,000 students through classroom instruction and national online learning platforms.
- Prepared students for competitive university admission examinations through structured lessons, practice materials, and academic planning.
- Translated advanced mathematical concepts into accessible problem-solving strategies for students with varied educational backgrounds.

SELECTED RESEARCH & PROJECTS

M.S. Thesis: Time-Series Analysis Using BART

R, Bayesian Modeling

- Conduct thesis research on the use of Bayesian Additive Regression Trees for time-series analysis under the supervision of Dr. Matthew T. Pratola.

Ordinary Least Squares from Scratch

Python, NumPy, Pandas, Matplotlib

- Develop an educational implementation of ordinary least squares regression from first principles, emphasizing matrix-based estimation and transparent statistical computation.

TECHNICAL SKILLS

Statistical Methods: Bayesian inference, BART, MCMC, hierarchical modeling, time-series analysis, linear and logistic regression, mixed-effects models, bootstrap methods, experimental design, and model selection

Statistical Learning: Ridge regression, lasso, decision trees, random forests, boosting, cross-validation, classification metrics, and predictive model evaluation

Programming: R, Python, C, C++, MATLAB, Bash, HTML, CSS, and JavaScript

Computing and Tools: Numerical optimization, algorithm benchmarking, code profiling, unit testing, Git, GitHub, LaTeX, RStudio, and VS Code

Languages: Bengali (native), English (fluent), Hindi (conversational), Urdu (conversational)

LEADERSHIP & HONORS

- Attended the 2026 Velocity Conference at UC Berkeley as an Indiana University representative for the science community.
- Collaborated on the development of a comprehensive online Linear Algebra curriculum during the COVID-19 pandemic.
- Co-organized the Australian Mathematics Competition in Bangladesh in 2014.